

OOP Principles, Interfaces & Collections Framework - Detailed Notes

DETAILED COMPREHENSIVE NOTES

Subject: Java Programming | Chapter: Chapter 2: Object-Oriented Programming & Collections | Academic Year: 2026-2027

1. The Four Pillars of Object-Oriented Programming

Encapsulation: Bundling data (attributes) and methods that operate on that data into a single unit (class), while restricting direct access using access modifiers (private, protected, public, package-private).

Inheritance: Mechanism where a child class acquires properties and behaviors of a parent class using the "extends" keyword. Java supports single class inheritance to avoid the diamond problem, but permits multiple interface implementation.

Polymorphism: Ability of an object to take many forms. Compile-time polymorphism is achieved through method overloading. Runtime polymorphism is achieved through dynamic method dispatch via method overriding with @Override.

Abstraction: Hiding internal implementation details and exhibiting only essential features using abstract classes and interfaces.

2. Abstract Classes versus Interfaces

Abstract classes can declare state (instance fields), constructors, and both abstract and concrete methods. They model an "is-a" relationship.

Interfaces define a contract. As of Java 8, interfaces can contain default and static methods. Java 9 introduced private interface methods. Functional interfaces contain exactly one abstract method and can be implemented via Lambda expressions.

- * A class can extend only one abstract class, but can implement unlimited interfaces.
- * Interface fields are implicitly public, static, and final.

3. Java Collections Framework Hierarchy

The Collections Framework provides a unified architecture for storing and manipulating groups of objects rooted at Iterable and Collection.

List: An ordered collection allowing duplicates. ArrayList provides O(1) random access backed by a dynamic array. LinkedList provides O(1) insertions at ends backed by a doubly-linked list.

Set: A collection containing no duplicate elements. HashSet provides O(1) average lookup using hash codes. TreeSet guarantees O(log n) ordered navigation using a Red-Black Tree.

Map: An object mapping keys to values with unique keys. HashMap uses an array of buckets with collision resolution via linked lists and balanced trees (TreeBins) when bucket capacity exceeds 8.